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EGYPT UNIVERSITY
OF INFORMATICS

Egypt University of Informatics

Faculty of Computing & Information Sciences

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Egyptian Real Estate Market Analysis

Identifying the Key Drivers of Property Prices in Egypt

Submitted by:

Razan Waleed — ID: 24-101157

Khaled Ahmed — ID: 24-101263

Youssef Mohamed — ID: 25-101800

Hamza Mohamed — ID: 24-101038

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Dataset: Egyptian Real Estate Listings — Kaggle

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1. Introduction

The Egyptian real estate market has changed a lot in recent years. Rapid urbanization, population growth, and economic shifts are driving these changes.

Property prices have gone up and down. This creates challenges and opportunities for buyers, sellers, and investors.

Knowing what affects property prices is very important. This study helps understand those key factors.

We analyzed a dataset of 19,925 real estate listings from Egyptian property websites. We looked at features like:

- Property size
- Number of bedrooms
- Location
- Property type

We used data cleaning, exploratory analysis, statistical tests, and machine learning. Our goal was to find what truly drives property prices in Egypt.

The findings in this report can help buyers, sellers, and real estate professionals make better decisions.

2. Research Question

At the heart of this project lies a single, focused research question:

“What are the key property features that significantly influence residential property prices in the Egyptian real estate market?”

To answer this question systematically, we break it down into five more specific sub-questions, each corresponding to a testable hypothesis:

1. Does property size (in square meters) have a significant positive relationship with price?
2. Does the number of bedrooms significantly affect price?
3. Are properties in Cairo significantly more expensive than those in other Egyptian cities?
4. Do Cash properties command significantly higher prices than Installment properties?
5. Are Villas significantly more expensive than Apartments?

Each of these questions is addressed through a dedicated statistical test in Section 7. Together, they paint a comprehensive picture of the Egyptian real estate pricing landscape.

3. Hypothesis

Based on domain knowledge and prior research into real estate markets, we formulate the following hypotheses:

Overall Hypothesis

H₀: There is no significant relationship between property features (area, bedrooms, location, payment method, type) and price.

H₁: Properties with larger area, more bedrooms, and premium types (e.g., Villas) have significantly higher prices in the Egyptian real estate market.

Sub-Hypotheses

#	Null Hypothesis (H ₀)	Alternative Hypothesis (H ₁)
H1	No significant relationship between size and price	Larger properties have significantly higher prices
H2	No significant difference in price across bedroom counts	More bedrooms leads to significantly higher prices
H3	No significant price difference: Cairo vs. other cities	Cairo properties are significantly more expensive
H4	No significant price difference: Cash vs. Installments	Cash properties are significantly more expensive
H5	No significant price difference: Villa vs. Apartment	Villas are significantly more expensive than Apartments

A significance level of $\alpha = 0.05$ is used for all tests. Any p-value below this threshold is considered statistically significant, providing sufficient evidence to reject the null hypothesis.

4. Dataset Overview

4.1 Data Source

The dataset used in this project — Egyptian Real Estate Listings — was sourced from Kaggle, contributed by Hassan Khaled. It consists of real-world property listings scraped from Egyptian real estate platforms, making it a highly relevant and authentic source for this analysis.

Attribute	Details
Dataset Name	Egyptian Real Estate Listings
Source	Kaggle — Hassan Khaled
Total Records	19,925 rows
Number of Features	12 columns
File Format	CSV
Data Nature	Real-world scraped listings

4.2 Feature Description

The dataset includes the following features, covering both numerical and categorical data types:

Feature	Type	Description
price	Numerical	Property price in Egyptian Pounds (EGP)
type	Categorical	Property type: Apartment, Villa, Chalet, Duplex, etc.
size	Text → Numerical	Property area in sqft and sqm (extracted sqm)
bedrooms	Text → Numerical	Number of bedrooms (including Maid room indicator)
bathrooms	Numerical	Number of bathrooms
payment_method	Categorical	Payment method: Cash or Installments

Feature	Type	Description
location	Text	Full property address (city extracted)
available_from	Date	Date the property becomes available
down_payment	Numerical	Down payment amount (72.7% missing — dropped)

5. Data Cleaning & Preprocessing

Raw data is rarely analysis-ready — and this dataset was no exception. Before drawing any conclusions, we invested significant effort in cleaning and transforming the data to ensure its quality and consistency. The preprocessing pipeline consisted of three main phases.

5.1 Removing Irrelevant Variables

Three columns were removed at the outset, as they either introduced unnecessary complexity or were too sparse to be useful:

- `url` — A unique identifier for each listing. It carries no predictive value for price and was therefore dropped.
- `description` — A free-text narrative for each property. While potentially rich in information, processing natural language would require advanced NLP techniques beyond the scope of this project.
- `down_payment` — This column had 14,479 missing values out of 19,925 entries (72.7% missing). Keeping it would have required extensive imputation that could introduce significant bias.

5.2 Handling Qualitative Variables

Each categorical variable required tailored treatment:

Property Type

Missing values were filled with the mode ('Apartment'). Rare property types with fewer than 100 occurrences were grouped under the label 'Other' to reduce noise. One-Hot Encoding was then applied to convert the categories into binary columns. Crucially, the original type column was preserved as `type_original` for use in hypothesis testing.

Bedrooms

This column contained entries like '3 Bedrooms + Maid' — a common format in Egyptian listings. We parsed these strings to extract the bedroom count and created a separate binary `maid_room` column (1 = has maid room, 0 = does not). Studio apartments were recoded as 1 bedroom.

Bathrooms

Inconsistent formatting (e.g., '3' and '3.0' for the same value) was standardized. The 'none' label was replaced with 0. All values were then converted to numeric format.

Payment Method

Rather than filling missing values with the mode — which would have artificially inflated one category — we used proportional random imputation, assigning missing values according to the existing Cash/Installments distribution. This approach preserves the natural balance of the data. The column was then label encoded, with Cash = 0 and Installments = 1. A readable payment_label column was retained for hypothesis testing.

Location

The location column contained full addresses such as 'Hyde Park, 5th Settlement, New Cairo City, Cairo'. We extracted the city name from the last segment of each address using a simple lambda function, creating a new city column. This was essential for the Cairo vs. Other Cities hypothesis test.

5.3 Handling Quantitative Variables

Size

The size column mixed square feet and square meters in a single string (e.g., '732 sqft / 68 sqm'). We used regular expressions to extract only the square meter value, which is the standard unit used in Egypt. Missing values were imputed with the median (a more robust measure than the mean, given the presence of outliers). Extreme outliers were then capped at the 5th and 95th percentiles to prevent them from distorting the analysis.

Price

Prices were stored as formatted strings with commas (e.g., '8,500,000'). We removed the commas and converted the column to float. Unlike size, rows with missing prices were dropped entirely — since price is our target variable, imputing it would be misleading. Outliers were similarly capped at the 5th and 95th percentiles.

5.4 Summary After Cleaning

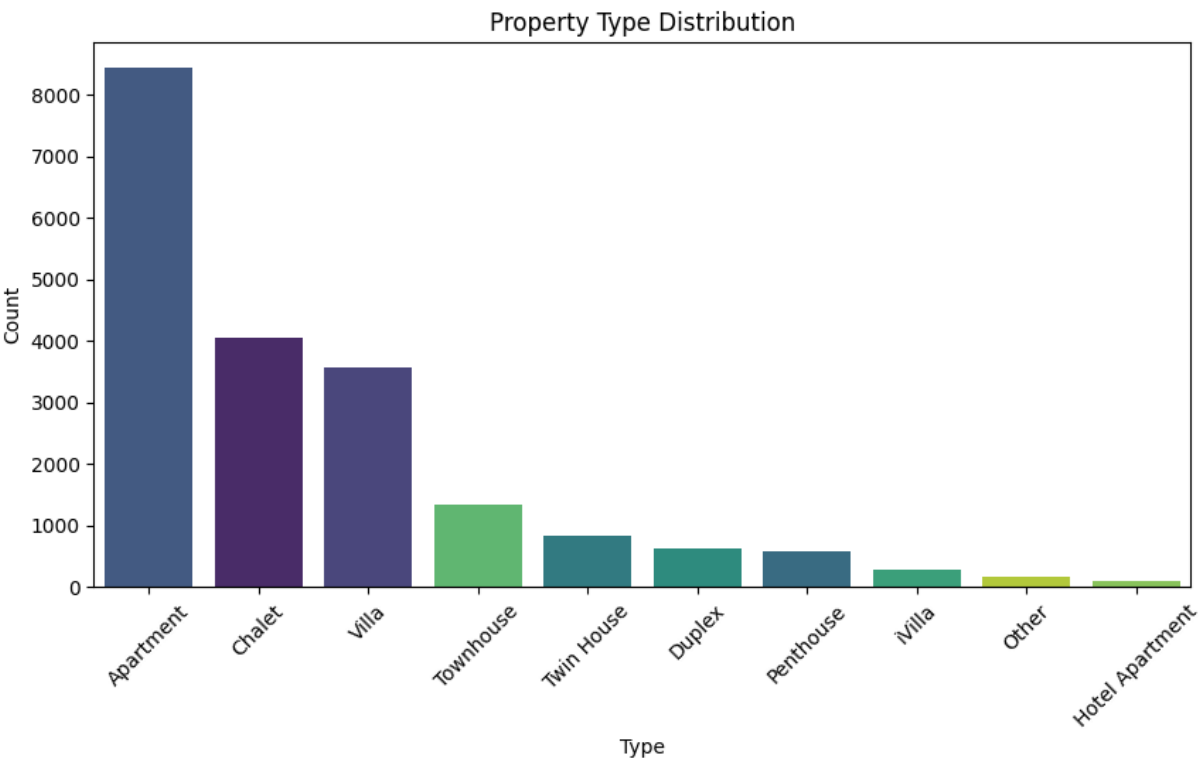
Metric	Before Cleaning	After Cleaning
Total Records	19,925	19,385
Missing Values	Multiple columns	0 (fully clean)
Columns	12	16 (engineered features added)
Price Range (EGP)	Various (with extremes)	500,000 — 48,750,000
Size Range (sqm)	Various (with extremes)	68 — 380 sqm

6. Exploratory Data Analysis

Before jumping into formal hypothesis tests, we conducted an exploratory data analysis (EDA) to understand the shape and distribution of the data, identify patterns, and generate intuitions that would inform our later analysis.

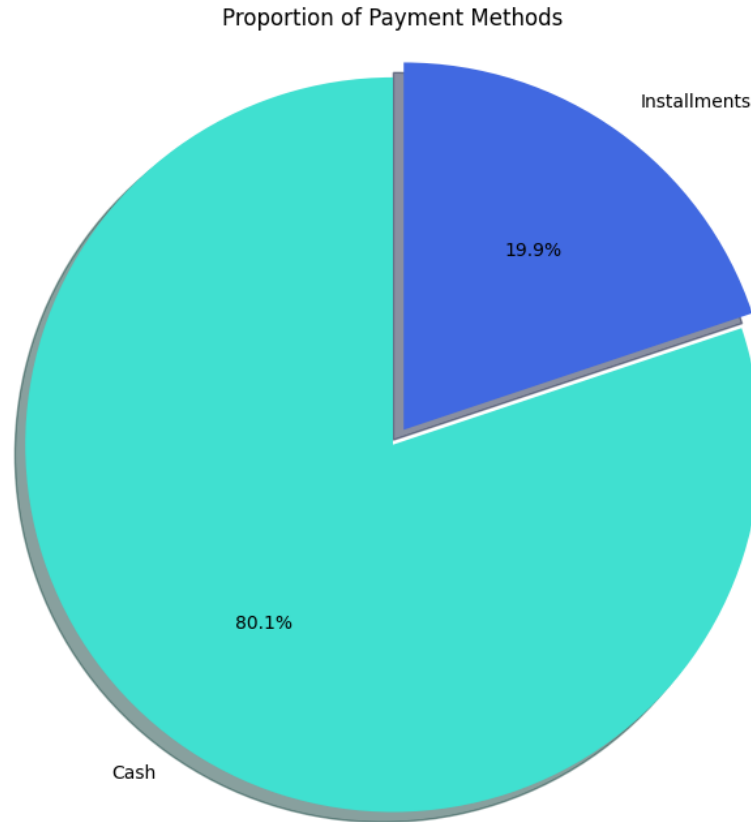
6.1 Distribution of Property Types

The countplot of property types reveals that Apartments dominate the listings, accounting for the vast majority of records. Villas, Chalets, and Duplexes each represent a smaller but notable share, while rarer types (Hotel Apartments, Townhouses, etc.) were grouped into 'Other'.



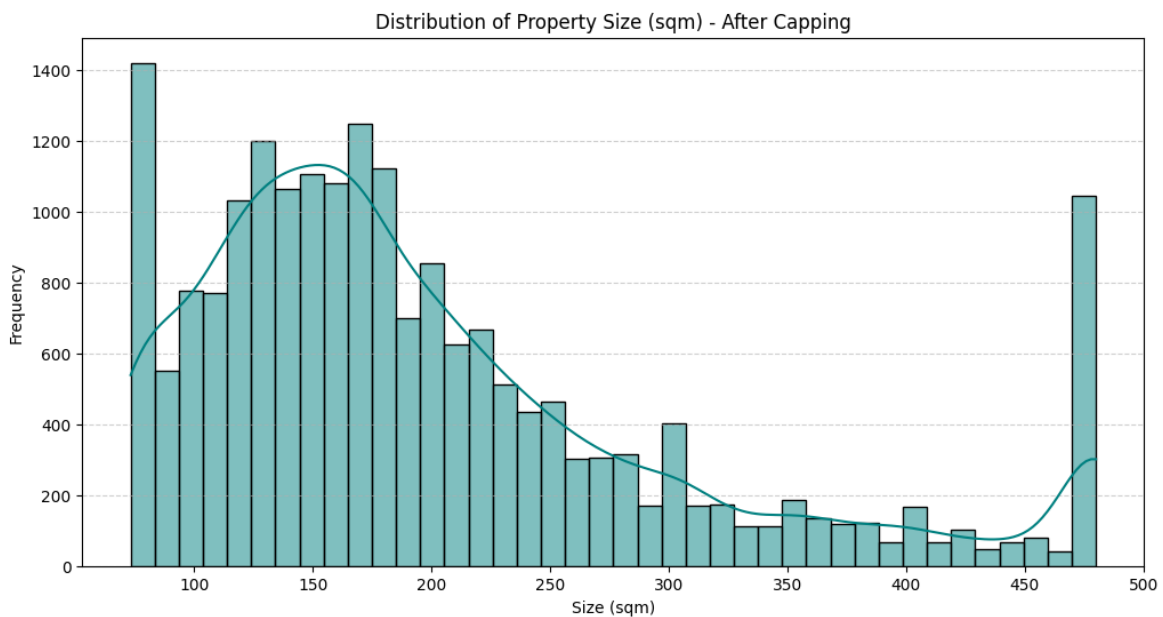
6.2 Payment Method Distribution

The pie chart of payment methods shows a clear skew: approximately 80% of listings are available for Cash purchase, while only 20% offer Installment options. This imbalance is worth noting when interpreting the Cash vs. Installments hypothesis test results.



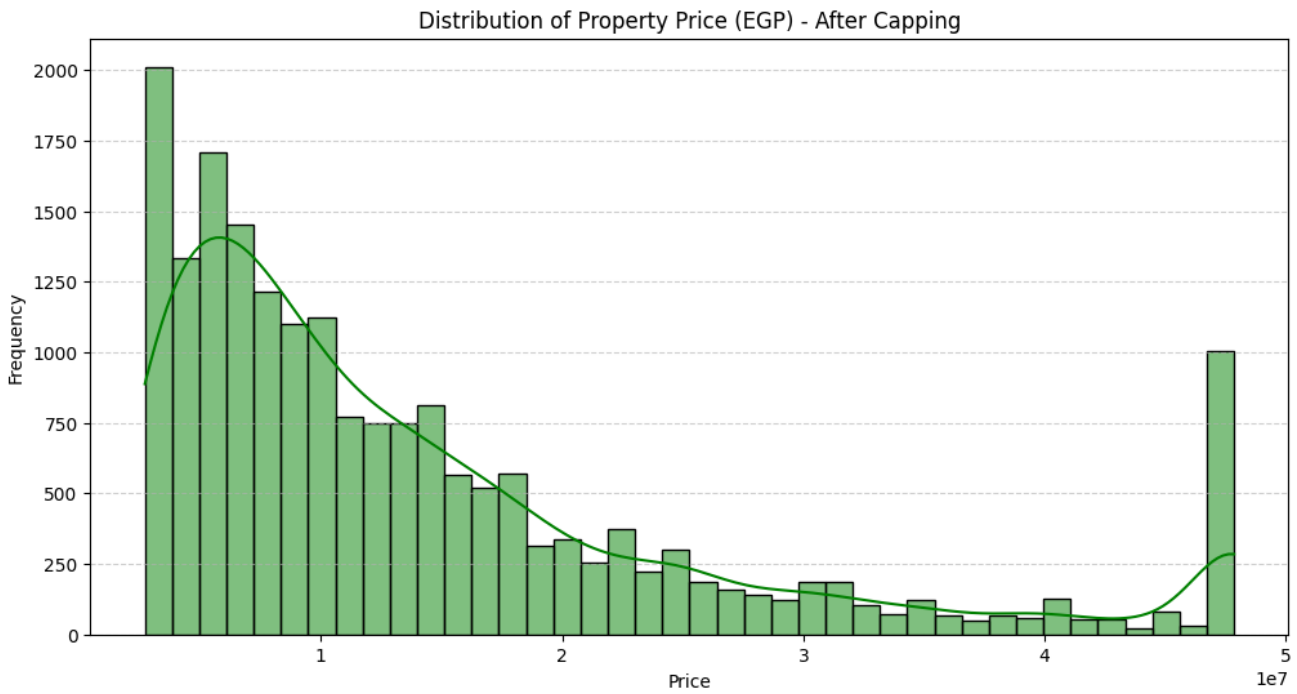
6.3 Property Size Distribution

After capping, the size distribution is right-skewed, with most properties falling between 120 and 200 sqm. The peak is around 150 sqm, which aligns with the typical Egyptian mid-market apartment. A long right tail indicates the presence of larger properties such as villas and duplexes.



6.4 Price Distribution

Similarly, the price distribution is right-skewed. The majority of listings are priced between 5 and 20 million EGP, with fewer high-end properties extending to the upper cap. This right-skewed distribution is typical of real estate markets worldwide.

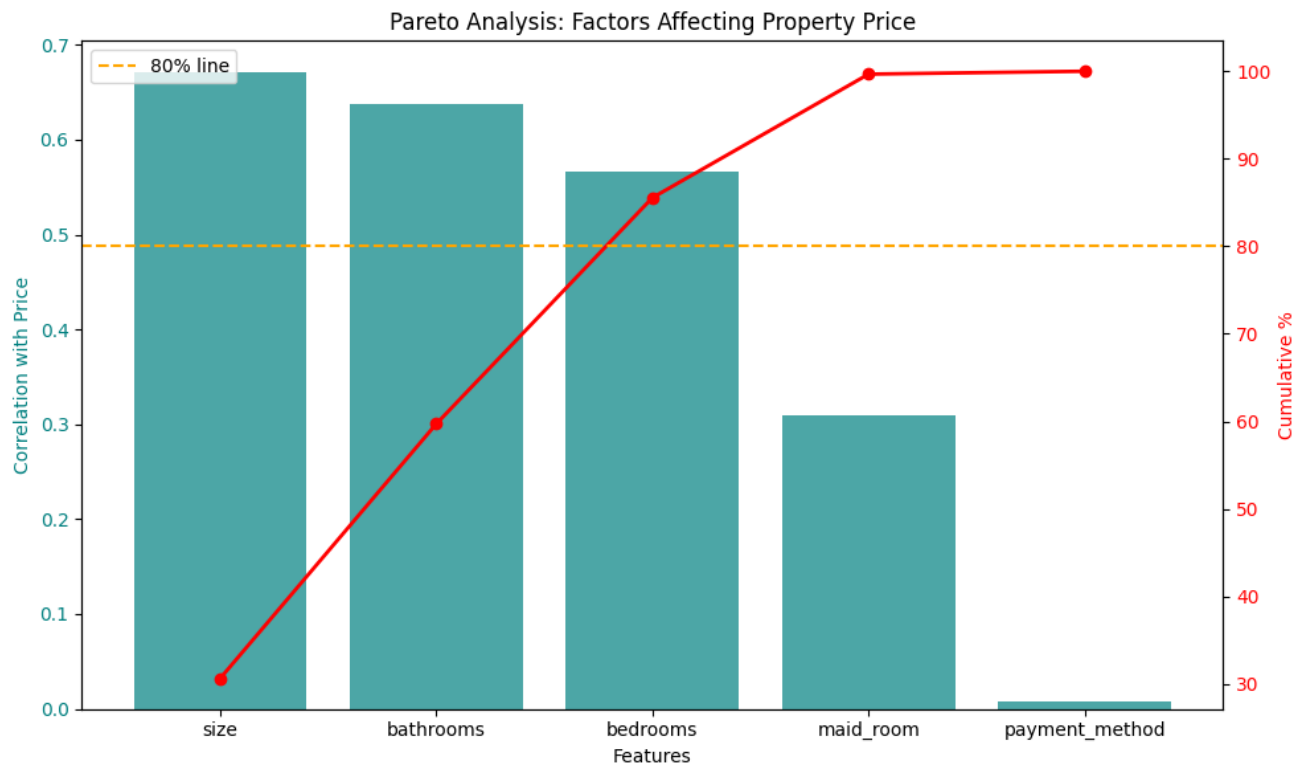


6.5 Pareto Analysis — What Drives Price?

A Pareto analysis was conducted to rank features by their absolute correlation with price. This analysis yielded one of the most insightful findings of the entire project:

Rank	Feature	Correlation with Price	Cumulative %
1	Size (sqm)	0.67	30%
2	Bathrooms	0.64	59%
3	Bedrooms	0.57	84%
4	Maid Room	0.31	98%
5	Payment Method	~0.00	100%

This table tells a compelling story: the top three features — size, bathrooms, and bedrooms — collectively explain over 80% of the price variation. Payment method, on the other hand, contributes virtually nothing. This finding directly informs both our hypothesis testing priorities and our feature selection for the machine learning model.



7. Hypothesis Testing

With a clear understanding of the data in hand, we proceed to formal hypothesis testing. Each hypothesis is addressed using the most appropriate statistical test, following the four-step hypothesis testing framework:

- 6. State the null and alternative hypotheses.
- 7. Choose the appropriate statistical test.
- 8. Calculate the test statistic and p-value.
- 9. Compare the p-value against $\alpha = 0.05$ and draw a conclusion.

To avoid artificially low p-values that arise from large sample sizes (a well-known issue with T-tests and ANOVA on datasets with thousands of rows), all tests were conducted on a random sample of $n = 500$ observations.

H1: Is There a Relationship Between Property Size and Price?

Test Used: Pearson Correlation Test — appropriate for examining the linear relationship between two continuous numerical variables.

Metric	Value
Correlation Coefficient	0.6707
P-Value	≈ 0.0000
Decision	Reject H_0

A correlation coefficient of 0.67 indicates a strong positive linear relationship between property size and price. This is one of the most intuitive findings of the study — larger properties simply cost more. Moreover, the p-value of approximately 0.0000 confirms that this relationship is not a statistical fluke; it is highly significant.



The scatter plot with a fitted regression line visually reinforces this finding, showing a clear upward trend as size increases. However, the spread of points around the regression line also suggests that size alone does not fully explain price — other factors are at play, which motivates the remaining hypotheses.

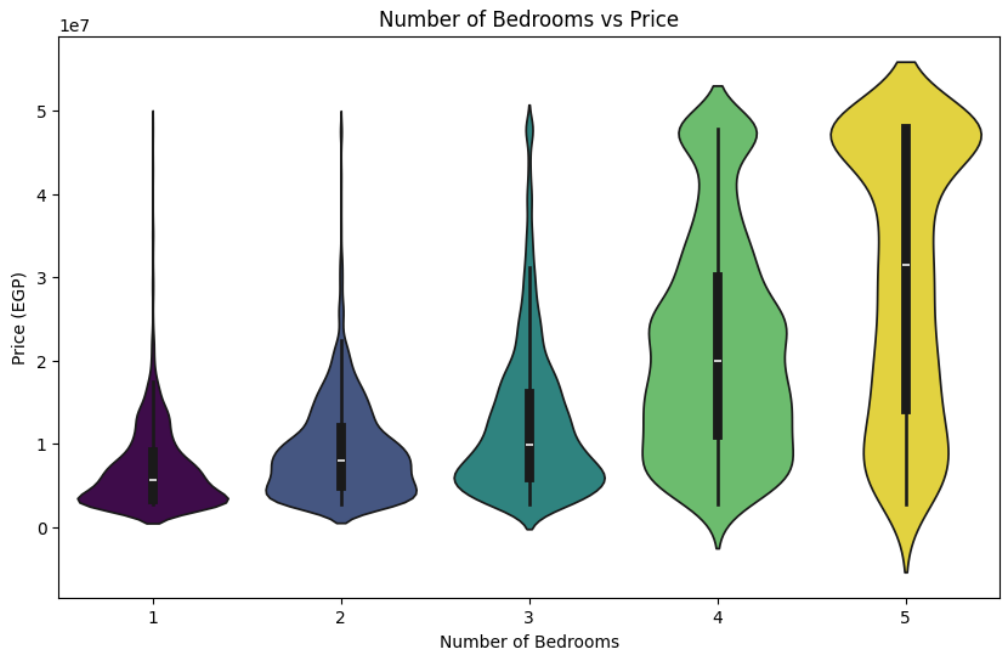
Conclusion: We reject H_0 . There is a strong, statistically significant positive relationship between property size and price.

H2: Does the Number of Bedrooms Affect Price?

Test Used: One-Way ANOVA — appropriate when comparing means across more than two groups. Here, the groups are properties with 1, 2, 3, 4, and 5 bedrooms.

Metric	Value
F-Statistic	1836.28
P-Value	0.0000
Decision	Reject H_0

The F-statistic of 1836.28 is extraordinarily large, indicating that the variance in price between bedroom groups far exceeds the variance within them. In practical terms, this means that the number of bedrooms is a powerful predictor of price.



The violin plot accompanying this test paints a vivid picture: as we move from 1-bedroom to 5-bedroom properties, the entire price distribution shifts upward. Furthermore, the widening of the violin shapes at higher bedroom counts reflects greater price variability among larger properties — understandable, since a 5-bedroom property could be anything from a mid-range apartment to a luxury villa.

Conclusion: We reject H_0 . The number of bedrooms has a highly significant effect on property price.

H3: Are Cairo Properties More Expensive Than Other Cities?

Test Used: Independent Samples T-test — appropriate for comparing the means of two independent groups (Cairo vs. all other cities).

Metric	Value
T-Statistic	-0.6858
P-Value	0.4932
Decision	Fail to Reject H_0

This was perhaps the most surprising result of the entire study. One might naturally assume that Cairo — as Egypt's capital and largest city — would command significantly higher property prices. However, the data tells a different story.



With a p-value of 0.4932 (well above the 0.05 threshold), we find no statistically significant difference between Cairo prices and those in other cities. The city distribution in our dataset offers an explanation: the second-largest group of listings (5,246 properties) comes from the North Coast, and a further 1,598 from the Red Sea. These are premium coastal destinations that attract high-end buyers, resulting in price ranges comparable to — or even exceeding — those in Cairo.

This finding challenges the conventional wisdom that Cairo always commands a price premium and highlights the growing importance of Egypt's coastal real estate market.

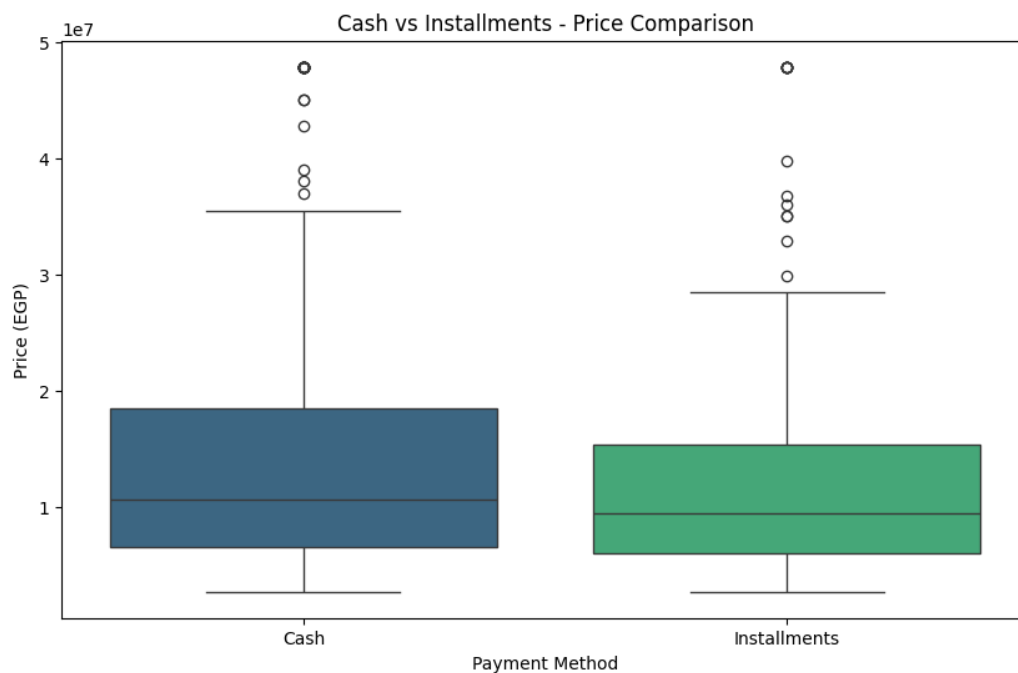
Conclusion: We fail to reject H_0 . Location (Cairo vs. other cities) does not have a statistically significant effect on price in this dataset.

H4: Do Cash Properties Cost More Than Installment Properties?

Test Used: Independent Samples T-test — appropriate for comparing the means of two independent groups (Cash vs. Installments).

Metric	Value
T-Statistic	1.2688
P-Value	0.2051
Decision	Fail to Reject H_0

Intuitively, one might expect Cash properties to be priced higher — either because developers charge a premium for full upfront payment, or because more expensive properties are less likely to offer installment plans. However, the data does not support this assumption.



With a p-value of 0.2051, we find no statistically significant difference in price between Cash and Installment properties. This result is consistent with the Pareto analysis, which showed that payment method has virtually no correlation with price. It suggests that in the Egyptian market, developers and sellers price properties based on physical characteristics (size, type, location) rather than payment terms.

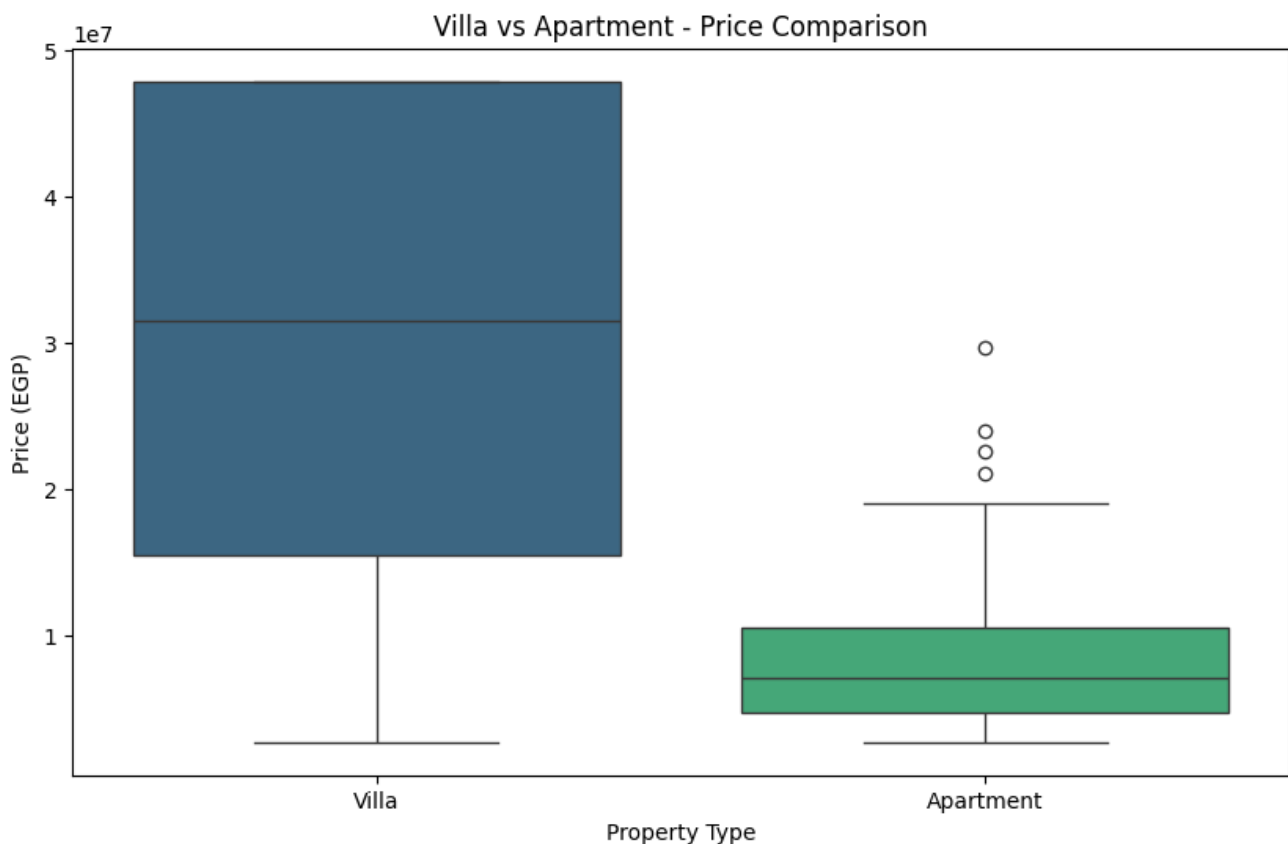
Conclusion: We fail to reject H_0 . Payment method does not significantly affect property price.

H5: Are Villas More Expensive Than Apartments?

Test Used: Independent Samples T-test — appropriate for comparing the means of two independent groups (Villas vs. Apartments).

Metric	Value
T-Statistic	18.2702
P-Value	0.0000
Decision	Reject H_0

Unsurprisingly, Villas are significantly more expensive than Apartments — and the margin is substantial. The T-statistic of 18.27 is one of the highest in our analysis, and the p-value of 0.0000 leaves no room for doubt.



The box plot for this test is particularly striking: the entire box for Villas sits far above the entire range of most Apartment prices. This reflects the fundamental difference between the two property types — Villas typically offer larger land plots, private outdoor spaces, and a level of luxury that Apartments simply cannot match at the same price point.

Conclusion: We reject H_0 . Villas are significantly and substantially more expensive than Apartments.

7.6 Hypothesis Testing Summary

#	Hypothesis	Test Used	Statistic	P-Value	Result
H1	Size vs Price	Correlation	$r = 0.6707$	~ 0.0000	Reject H_0
H2	Bedrooms vs Price	ANOVA	$F = 1836.28$	0.0000	Reject H_0
H3	Cairo vs Others	T-test	$t = -0.6858$	0.4932	Fail to Reject H_0
H4	Cash vs Installments	T-test	$t = 1.2688$	0.2051	Fail to Reject H_0
H5	Villa vs Apartment	T-test	$t = 18.2702$	0.0000	Reject H_0

8. Machine Learning Model — Linear Regression

Building on the insights from our hypothesis testing, we developed a Linear Regression model to predict property prices. Linear Regression is particularly well-suited for this task: it is interpretable, computationally efficient, and directly quantifies the relationship between each feature and price.

8.1 Feature Selection

Rather than feeding all available columns into the model, we made deliberate, theory-driven choices about which features to include — guided directly by the hypothesis testing results and Pareto analysis:

Feature	Included?	Reason
size	Yes	Strongest predictor ($r = 0.67$), H1 confirmed
bedrooms	Yes	Highly significant (ANOVA, $p < 0.05$), H2 confirmed
bathrooms	Yes	2nd highest Pareto ranking
maid_room	Yes	Moderate impact in Pareto analysis
Property type (encoded)	Yes	Significant impact confirmed by H5
city	No	H3 failed — no significant location effect
payment_method	No	H4 failed — negligible correlation with price

This is a key strength of the study: the machine learning model is not built on guesswork. Instead, every feature inclusion and exclusion is backed by statistical evidence. Features that showed no significant relationship with price were deliberately left out — a principle known as parsimony — to avoid overfitting and keep the model clean and interpretable.

8.2 Model Training

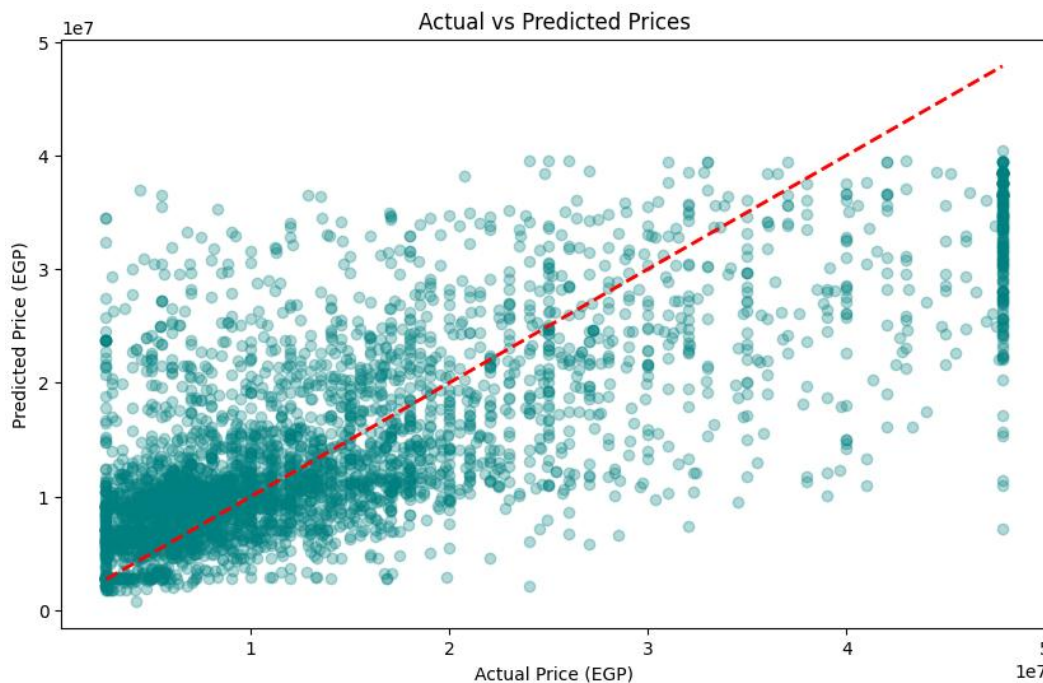
The dataset was split into a training set (80%) and a test set (20%), with a fixed random state of 42 for reproducibility.

Parameter	Value
Model	Linear Regression (sklearn)
Training Set	15,508 records (80%)
Testing Set	3,877 records (20%)
Features Used	12 (including One-Hot encoded type columns)
Random State	42

8.3 Model Performance

Metric	Value	Interpretation
R ² Score	0.5344	Model explains 53% of price variance
MSE	64,043,433,397,188 EGP ²	Mean squared prediction error
RMSE	8,002,714 EGP	Average prediction error ≈ 8 million EGP

An R^2 of 0.53 means the model explains just over half of the variation in property prices using only five categories of features. This is a respectable result for a Linear Regression model on real-world data, particularly given that the remaining 47% of variance is likely driven by factors not present in our dataset — such as floor level, finishing quality, proximity to schools and transport, and developer reputation.



The scatter plot of Actual vs. Predicted Prices shows that the model performs best for mid-range properties (the cluster of points closest to the red dashed perfect-prediction line). For higher-priced properties, the model tends to underestimate — a common challenge in real estate regression, where luxury properties often command prices driven by subjective factors that no model can easily capture.

The vertical cluster of points on the right side of the scatter plot is an artifact of the price capping applied during preprocessing: since all prices above the 95th percentile were capped at the same value, many high-end predictions are mapped to the same point.

8.4 Price Predictor

To make the model tangible and practical, we implemented a simple price prediction function. A user can input property characteristics — size, number of bedrooms, bathrooms, maid room availability, and property type — and receive an estimated price in EGP. This tool demonstrates the real-world utility of the analysis beyond academic hypothesis testing.

9. Conclusion

This project set out to answer a straightforward but commercially important question: what drives property prices in Egypt? After analyzing nearly 20,000 real estate listings using a rigorous combination of data cleaning, exploratory analysis, hypothesis testing, and machine learning, we are now in a position to answer that question with confidence.

The most important takeaways from this study are:

- Property size is the strongest driver of price, showing a clear and statistically significant positive relationship with property value.
- The number of bedrooms has a major impact on price, with larger properties consistently commanding higher prices.
- Property type plays a critical role, as Villas are significantly more expensive than Apartments due to their larger size and premium features.
- Location (Cairo vs. other cities) does not have a significant effect on price, highlighting the rising importance of coastal and luxury areas outside Cairo.
- Payment method (Cash vs. Installments) has no meaningful impact on property prices, confirming that pricing is driven by property characteristics rather than transaction terms.

Together, these findings confirm that the Egyptian real estate market is primarily driven by physical property characteristics (size, bedrooms, type) rather than transactional or geographic factors. This knowledge empowers buyers to focus on what truly drives value, and helps sellers position their properties more effectively.

The Linear Regression model, while not perfect ($R^2 = 0.53$), provides a solid foundation for price estimation. Its 47% unexplained variance is not a failure — it is a reminder that real estate prices are partly driven by factors that no dataset can fully capture: the charm of a particular view, the quality of a finish, or the reputation of a developer.

10. Future Improvements

While this study provides robust findings, several avenues for improvement could enhance both the accuracy and scope of the analysis in future work:

10. Enrich the feature set. Adding variables such as floor level, finishing quality (fully finished vs. semi-finished vs. core & shell), proximity to amenities (metro stations, schools, malls), and developer reputation would likely push the R^2 significantly higher.
11. Try advanced ML models. Random Forest, Gradient Boosting (XGBoost), or Neural Networks could capture non-linear relationships that Linear Regression misses, potentially delivering substantially better predictive accuracy.
12. Granular geographic analysis. Instead of comparing Cairo vs. other cities as a binary split, future analysis could compare specific districts and neighborhoods (e.g., New Cairo vs. Zamalek vs. Sheikh Zayed) to uncover more nuanced location effects.
13. Time series analysis. Adding a temporal dimension — tracking how prices have changed over time — would allow for trend analysis and forecasting, making the model far more useful for investment decisions.
14. NLP on property descriptions. The description column, which we dropped for simplicity, likely contains valuable signals (e.g., 'sea view', 'fully furnished', 'prime location') that could be extracted and used as features.

11. References

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